

Physics Lecture Notes

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Chapter 1

Introduction

These notes present selected foundations of physics with an emphasis on mathematical structure, conceptual clarity, and model-based reasoning. The course starts from mathematical tools and then develops the language of motion, forces, oscillations, waves, electromagnetic phenomena, measurements, cosmology, relativity, and modern physics.

Physics uses mathematics to describe patterns in nature, but the meaning of a formula depends on the assumptions behind the model. For this reason, the notes aim to combine equations with interpretation: what quantities represent, how they are measured, when a model applies, and how different parts of the theory fit together.

Chapter 2

Mathematical Foundations

2.1 Introduction

This section provides an introduction to the mathematical foundations of physics.

Chapter 3

Kinematics

3.1 Introduction

This section provides an introduction to kinematics.

Chapter 4

Dynamics

4.1 Introduction

This section provides an introduction to dynamics.

Chapter 5

Oscillations and Waves

5.1 Introduction

This section provides an introduction to oscillations and waves.

Chapter 6

Electromagnetism

6.1 Introduction

This section provides an introduction to electromagnetism.

Chapter 7

Statistics and Measurements

7.1 Introduction

This section provides an introduction to statistics and measurements.

Chapter 8

Cosmology

8.1 Introduction

This section provides an introduction to cosmology.

Chapter 9

Relativity

9.1 Introduction

This section provides an introduction to relativity.

Chapter 10

Modern Physics

10.1 Introduction

This section provides an introduction to modern physics.